PERT

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Program Evaluation and Review Technique (PERT)

- PERT is a project management technique, where planning, scheduling, organizing, coordinating and controlling uncertain activities are done.
- PERT is a probabilistic estimate of the randomness of project events, and unlike the deterministic CPM technique, it can also be used to predict which of the scheduled tasks are likely to slip.
- PERT was introduced by the consulting firm “Booz-Allen and Hamilton” in the 1950s to satisfy the demands of the U.S. Navy Special Projects Office.

Framework of PERT

- Similar to CPM, PERT follows six basic steps:
- Define the project and prepare the work breakdown structure.
- Develop the relationships among the activities. Decide which activities must precede and which must follow others.
- Draw the network connecting all the activities.
- Assign time and/or cost estimates to each activity.
- Compute the *longest* time path through the network. This is called the **critical path**.
- Use the network to help plan, schedule, monitor, and control the project.

Variability in Activity Times

- In identifying all earliest and latest times for any activity and the associated critical path(s), the CPM approach has been adopted.
- It is assumed that all activity times are known and fixed constants.
- However, in practice, activity completion times vary depending on various factors.
- For example, say activity A requires raw materials to be procured. Its activity time is 3 weeks. If raw material is not procured at the proper time due to some reason, the activity will be delayed.

- In practice, some activities are less prone to delay while other activities are highly prone to delay.
- The variabilities in activity time cannot be ignored while deciding the schedule of any project. PERT addresses this issue.

Some Differences between PERT and CPM

Basis of Comparison	PERT	CPM
What is it?	A technique using probabilistic approach	A technique using deterministic approach
Orientation	Event-oriented	Activity-oriented
Role in crashing	Limited	Extensive
Estimates	Three time estimates	One time estimates
Focuses on	Time	Time-cost trade-off
Management of	Unpredictable activities	Predictable activities

https://keydifferencs.com/difference-between-pert-and-cpm.html

Three time estimates in PERT

- In PERT, a probability distribution based on three time estimates for each activity is employed as follows:

  - **Optimistic time (t_o)** = time an activity will take if everything goes as planned. In estimating this value, there should be only a small probability (say, 1/100) that the activity time will be $< t_o$.

  - **Pessimistic time (t_p)** = time an activity will take assuming very unfavorable conditions. In estimating this value, there should also be only a small probability (also 1/100) that the activity time will be $> t_p$.

  - **Most likely time (t_m)** = most realistic estimate of the time required to complete an activity.

Three time estimates in PERT

The expected activity time (t_e) is calculated as

$$t_e = \frac{(t_p + 4t_m + t_o)}{6}$$

- The most likely time (t_m) is given four times the weight as the optimistic time (t_o) and pessimistic time (t_p).
- The time estimates computed for each activity is used in the project network to compute all earliest and latest times.
- The project completion time is the sum of expected times (t_e) for all activities on the critical path.

PERT

To compute the dispersion or the variance of the activity completion time:

$$\text{Variance} = \left\{ \frac{t_p - t_o}{6} \right\}^2$$

***Example*:** Compute the expected time and variance for the project. The values of t_o, t_m and t_p are provided.

Activities	Optimistic time (t_o)	Most likely (t_m)	Pessimistic time (t_p)	Expected time (t_e)	Variance
A	2	3	4	3	0.11
B	1	2	3	2	0.11
C	2	3	10	4	1.78
D	1	2	3	2	0.11
E	3	4	5	4	0.11
F	2	3	4	3	0.11
G	3	5	13	6	2.78
H	1	2	3	2	0.11

$$t_e = \frac{(t_p + 4t_m + t_o)}{6}$$

$$\text{Variance} = \left\{\frac{t_p - t_o}{6}\right\}^2$$

- In some activities, the variance is relatively higher.
- Variance gives information about the degree of spread in the data set.

Probability of Project Completion

- The critical path analysis assisted in determining that the project completion time is 15 weeks. However, there is significant variation in the time estimates for several activities.
- Variation in activities that are on the critical path can delay the overall project completion time.
- PERT uses the variance of critical path activities to help determine the variance of the overall project. Project variance is computed by summing variances of critical activities:

$$\text{Project variance} = \sigma_p^2 = \sum(\text{Variances of activities on critical path})$$

Some assumptions on PERT analysis

- Assumptions of PERT are
  1. total project completion times follow a normal probability distribution
  2. activity times are statistically independent.

- With these assumptions, the bell-shaped normal curve can be drawn to represent the project completion dates.
- The normal curve implies that there is a 50% chance that the manufacturer's project completion time will be less than 19 weeks and a 50% chance that it will exceed 19 weeks.

Probability of Project Completion

- ***Example***: Evaluate the project variance and standard deviation of the project.

Solution: The variance of project is based on the critical path (Start–A–C–E–G–H). The project variance is

$$\text{Project variance}\left(\sigma_p^2\right) = 0.11 + 1.78 + 0.11 + 2.78 + 0.11 = 4.89$$

The standard deviation of the project is

$$\text{Standard deviation}\left(\sigma_p^2\right) = \sqrt{\text{Variance}} = \sqrt{4.89} = 2.21 \text{ weeks}$$

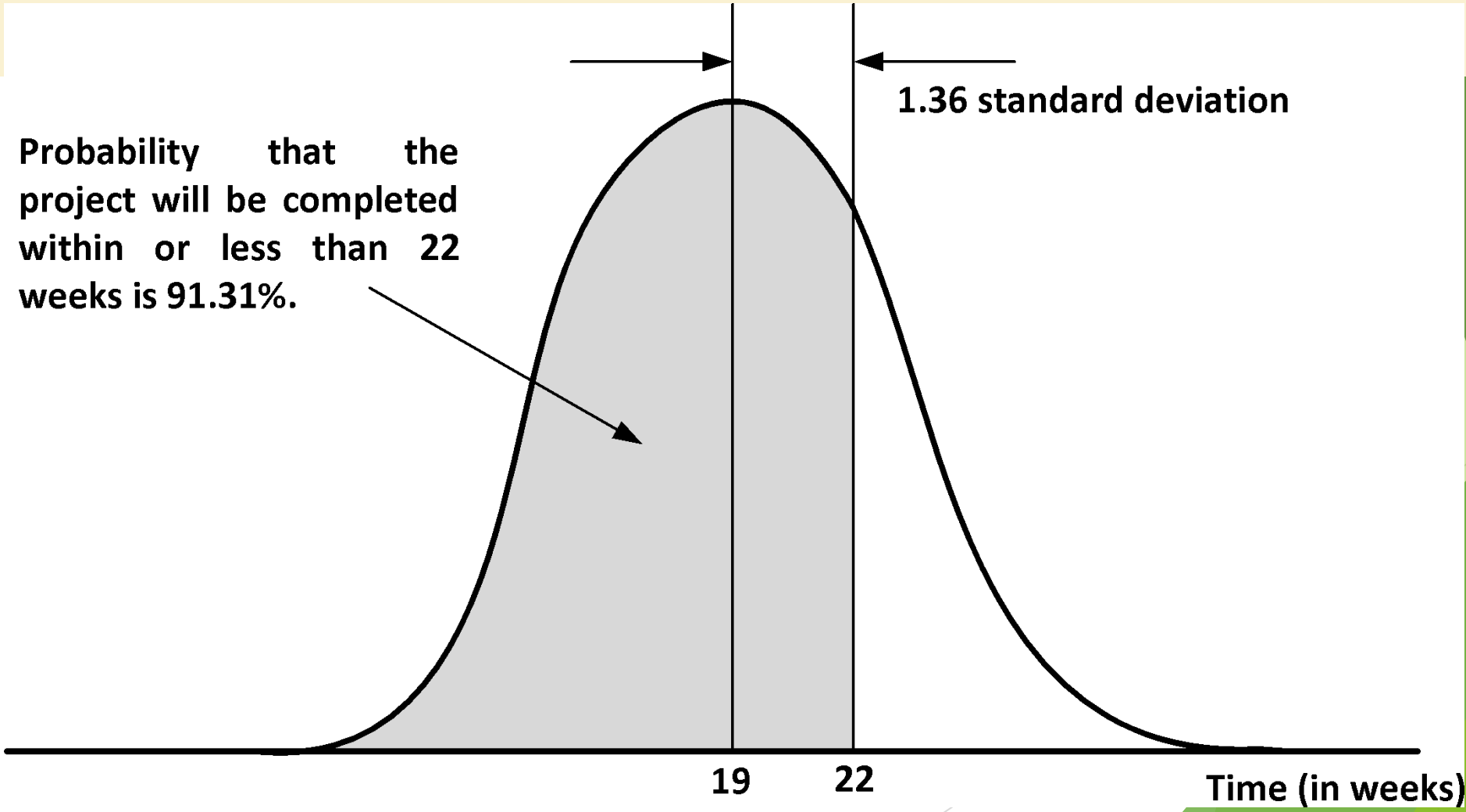
Activities	Expected time (t_e)	Variance
A	3	0.11
B	2	0.11
C	4	1.78
D	2	0.11
E	4	0.11
F	3	0.11
G	6	2.78
H	2	0.11

- ***Example:*** The deadline (due date) of the project is 22 weeks. What is the probability that the project will be completed before the due date?
- ***Solution:*** The expected completion date of the project is 19 weeks. Standard deviation is 2.21 weeks. The standard normal equation is given by

$$Z = \left(\text{Due date} - \text{Expected date of completion}\right)/\sigma_p$$
$$= \left(22 - 19\right)/2.21$$
$$= 1.36$$

where *Z* is the number of standard deviations the due date/target date lies from the mean/expected date.

- Referring to the *Z*-table or using **NORMSDIST (1.36)** command in MS excel, the probability is obtained as 0.913085.
- Hence, there is a probability of 91.31% that the project will completed within 22 weeks or less.



- ***Example:*** Instead of 91.31% probability, what should be the due date/deadline so that the probability of completion of the project becomes 99% ?
- ***Solution:*** Referring to the Z-table or using **NORMSINV (0.99)** command in MS excel, the Z value is obtained as 2.3263. Using the standard normal equation, the due date is estimated as follows:

$$Z = \left(\text{Due date} - \text{Expected date of completion}\right)/\sigma_p$$

$$\text{or Due date} = \text{Expected date of completion} + Z\sigma_p$$

$$\text{or Due date} = 19 + 2.3263 \times 2.21 = 24.14 \text{ weeks}$$